

PK-Net: A prior knowledge-driven dual-path network for enhanced glaucoma screening

Xiaoyan Kui^a, Zeru Hai^{a,b,*}, Bei Zou^{a,b}, Yang Li^b, Wei Liang^c, Zuheng Ming^d, Liming Chen^e

^a School of Computer Science and Engineering, Central South University, 410083, Changsha, Hunan, China

^b School of Informatics, Hunan University of Chinese Medicine, 410208, Changsha, Hunan, China

^c School of Computer Science and Engineering, The Xiangjiang Laboratory, Hunan University of Science and Technology, 411199, Xiangtan, Hunan, China

^d Laboratoire L2TI, Institut Galilée, Université Sorbonne Paris Nord (USPN), 99 Avenue Jean-Baptiste Clément, 93430, Villetaneuse, France

^e Ecole Centrale de Lyon University of Lyon, Lyon, France

ARTICLE INFO

Keywords:

Glaucoma screening
Prior knowledge-driven
Optic disc region
Binocular relationship

ABSTRACT

Glaucoma is a chronic ocular disease that often remains undiagnosed until advanced stages, highlighting the need for early detection. Current state-of-the-art methods mainly adapt attention mechanisms into classification networks designed for natural images via transfer learning, but they fail to capture domain-specific features and show weak cross-dataset generalization. In this paper, we propose a prior knowledge-driven dual-path network (PK-Net) that integrates medical knowledge into model architecture for glaucoma screening. First, based on the diagnostic importance of the optic disc, we introduce the Global and Local Fusion Network (GloLocNet), which combines high-resolution local optic disc images with global fundus images and applies a triple-loss strategy to improve feature extraction. Second, to leverage the strong inter-eye correlation of glaucoma, we propose the Binocular Fusion Network (BFNet), where paired eye images are processed through parallel GloLocNet encoders and fused to yield joint screening results. PK-Net was validated on multiple datasets, achieving superior intra- and cross-dataset results. For PAPILA, AUC, BAcc, Sen, and Spe were 95.67 %, 91.18 %, 88.72 %, and 93.63 %; for OIA-ODIR, 92.43 %, 85.64 %, 84.62 %, and 86.67 %. Trained on ORIGA and tested on REFUGE, the values were 90.00 %, 82.29 %, 81.25 %, and 83.33 %, demonstrating strong generalization. On GAMMA, results were 95.16 %, 83.37 %, 75.49 %, and 91.29 %. These findings indicate that PK-Net effectively enhances glaucoma screening by embedding prior medical knowledge into network design.

1. Introduction

Glaucoma is a prevalent and difficult-to-treat eye disease that stands as the second leading cause of blindness globally, earning the moniker “silent killer of vision [1]”. This condition can affect individuals of any age, with an overall prevalence ranging from 2.0 % to 2.8 % [2]. The risk of glaucoma increases with age, affecting about 4 % to 5 % of people by the age of 50, and rising to 9.2 % in those over 70 [3]. The most common symptom is the progressive loss of visual fields, which may initially present as dark spots or blurred vision. Unfortunately, by the time these symptoms are noticeable, the disease has often progressed to a more advanced stage, making it challenging to restore lost vision. Currently, color fundus digital imaging is widely utilized in clinical settings for glaucoma screening [4]. Ophthalmologists typically assess fundus images to diagnose glaucoma by examining certain indicators [5], such as an increased cup-to-disc ratio (CDR), optic nerve damage, and elevated

intraocular pressure (IOP). Compared to the complex measurements required for IOP, evaluating the optic nerve head or CDR is a simpler method for early glaucoma detection. Since CDR is closely related to IOP, an increase in IOP often results in optic nerve head damage, which may be accompanied by an increased CDR. Generally, a normal CDR value is less than 0.2 [6]. When the CDR falls between 0.2 and 0.5, the likelihood of glaucoma increases and a CDR greater than 0.6 may signal the early stages of glaucoma or other ocular diseases, necessitating further investigation to determine whether the cause is pathological.

However, the limitations of manual screening underscore the urgent need for automated approaches [7]. Manual screening is inherently time-consuming and heavily reliant on the expertise of ophthalmologists. It may lead to inconsistencies in diagnosis, particularly in regions where specialized care is unavailable [8]. The rising prevalence of glaucoma, driven by an aging population, exacerbates the demand for ophthalmic services, further straining already scarce resources [9].

* Corresponding author.

E-mail address: haizeru@163.com (Z. Hai).

<https://doi.org/10.1016/j.knosys.2025.114374>

Received 30 November 2024; Received in revised form 16 August 2025; Accepted 28 August 2025

Available online 30 August 2025

0950-7051/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Automated screening systems have the potential to standardize and accelerate the diagnostic process, alleviating the burden on healthcare providers and reducing the costs associated with traditional screening methods [10]. By integrating advanced deep learning techniques, these systems can accurately detect and evaluate critical indicators of glaucoma, such as CDR and optic nerve damage, thereby enabling early detection and timely intervention [11].

Despite notable progress in automated glaucoma screening, most existing deep learning methods still follow general-purpose design strategies, such as using attention mechanisms or multi-path inputs. However, these methods show limited improvement in feature learning and classification performance for glaucoma. In clinical practice, glaucoma diagnosis often relies on prior medical knowledge, including changes in the optic disc structure such as an increased cup-to-disc ratio, and the strong symmetry between the left and right eyes. Current models do not make full use of this knowledge, which limits their ability to focus on key regions and to learn cross-eye feature relationships. To address this limitation, we propose PK-Net, a prior knowledge-driven dual-path network specifically designed for glaucoma screening. Our method explicitly incorporates the critical diagnostic role of the optic disc and the pathological correlation between both eyes to develop a deep learning model tailored for glaucoma detection. By embedding domain-specific medical knowledge into the network architecture, PK-Net enhances screening accuracy and robustness, outperforming conventional deep learning approaches.

2. Related work

2.1. Glaucoma diagnostic methods

Intelligent diagnostic methods for glaucoma generally fall into two categories: image segmentation and image classification. In image segmentation, the optic cup and disc areas are first segmented, and the cup-to-disc ratio (CDR) is calculated to diagnose glaucoma. These methods allow for precise numerical calculations but are highly dependent on the accuracy of the segmentation. For example, Fu et al. proposed a multi-label joint segmentation deep network for discs and cups, followed by CDR calculation to validate glaucoma detection results [12]. However, these measurement-based methods are vulnerable to inaccuracies, particularly when dealing with pathological areas or low-contrast images, which can introduce noise into the analysis.

The image classification methods bypass the segmentation step by directly extracting features from fundus images and predicting glaucoma or normal eye conditions. These methods can be divided into machine learning and deep learning approaches. Machine learning methods have been applied in various prediction tasks, demonstrating their ability to model complex patterns [13–15]. However, they rely heavily on manual feature extraction and predefined assumptions, limiting their adaptability to complex glaucoma screening tasks. For instance, Rudiger et al. developed a two-stage classification scheme based on fundus structure, where features are extracted and a support vector machine (SVM) is used

for classification [16]. Similarly, Koh et al. employed accelerated robust feature descriptors and histograms of oriented gradients for glaucoma classification [17]. However, these traditional methods require expert knowledge in feature selection and exhibit poor adaptability to variations in image quality or complex lesions [18–20].

With the rapid advancement of deep learning, various neural network architectures have been widely applied, demonstrating their effectiveness in capturing complex patterns [21–23]. These advancements further reinforce the motivation for leveraging deep learning in medical image-based glaucoma screening [24–26]. Fu et al. introduced the Disc-aware Ensemble Network (DENet) for automatic glaucoma screening, integrating global image analysis, disc segmentation, and local disc region processing [27], as shown in Fig. 1(b). Despite its comprehensive approach, DENet's large model size and reliance on disc localization and segmentation present challenges, especially when working with datasets lacking segmentation labels. Das et al. proposed CA-Net, which incorporates cascaded attention mechanisms to enhance feature learning, but it introduces attention only in the final stages, without specific adaptations for glaucoma [28]. Other models like GCNet [29], GS-Net [30] and AES-Net [31] have also integrated attention mechanisms. However, these mechanisms are typically applied in the later stages of the network, specifically just before the final average pooling layer, as illustrated in Fig. 1(a). Recently, the DeepGD model [32] was introduced, which is based on EfficientNet-B4 combined with a Snapshot Ensemble strategy (Cosine Cyclic Learning Rate Schedule for Snapshot Ensemble CNN) for glaucoma screening. However, these models are not specifically adapted for glaucoma screening, which may limit their generalizability and effectiveness in this particular task.

In summary, while attention mechanisms have been effective in improving feature learning, their implementation predominantly in the later stages of the network and their general design limit their impact on glaucoma screening. These models lack glaucoma-specific methodologies that address the unique challenges of the disease.

2.2. Binocular relationship

The relationship between the left and right eye is significant in many ocular diseases, where similar conditions often affect both eyes simultaneously. For example, myopia, hyperopia, and astigmatism are usually present in both eyes with similar refractive errors [33]. This bilateral relationship extends to other ocular diseases like diabetic retinopathy (DR), which often manifests symmetrically in both eyes due to high blood glucose levels [34]. Studies indicate that over 95 % of patients in the EyePACS DR dataset show consistent lesion scores in both eyes, highlighting the potential of leveraging bilateral relationships to improve diagnostic accuracy [35].

Glaucoma also exhibits a notable bilateral relationship. In China, approximately 21 million people suffer from glaucoma, with 5.67 million experiencing bilateral blindness [36]. While glaucoma may initially affect one eye more severely, the other eye typically becomes involved as the disease progresses. This emphasizes the importance of early detection and monitoring of both eyes to slow down optic nerve damage and vision loss [37]. Kovalyk et al. suggested using bilateral data to develop algorithms that leverage joint information from both eyes, improving early glaucoma diagnosis [38]. Relying on data from just one eye may miss early symptoms, making the use of bilateral images crucial.

Therefore, using bilateral fundus images in glaucoma screening is essential. By utilizing the complementary information from both eyes, models can gain more robust insights during training. When image quality differs between the eyes, the better-quality image can enhance the analysis, ultimately improving screening accuracy and model generalization, even with partially poor-quality images. Based on previous glaucoma screening models and combined with high binocular correlation of glaucoma, we propose a model specifically designed for glaucoma screening, as shown in Fig. 1(c).

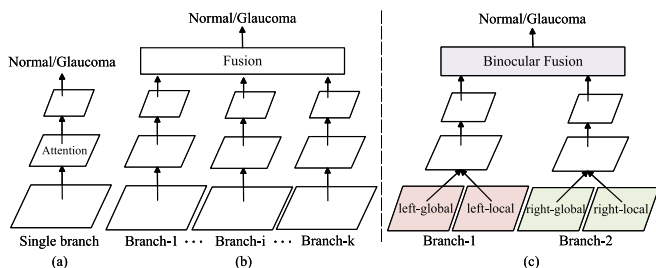


Fig. 1. Comparison of different CNN architectures for glaucoma screening (a) Embedded attention mechanism. (b) Multiple branch groups. (c) Our proposed PK-Net.

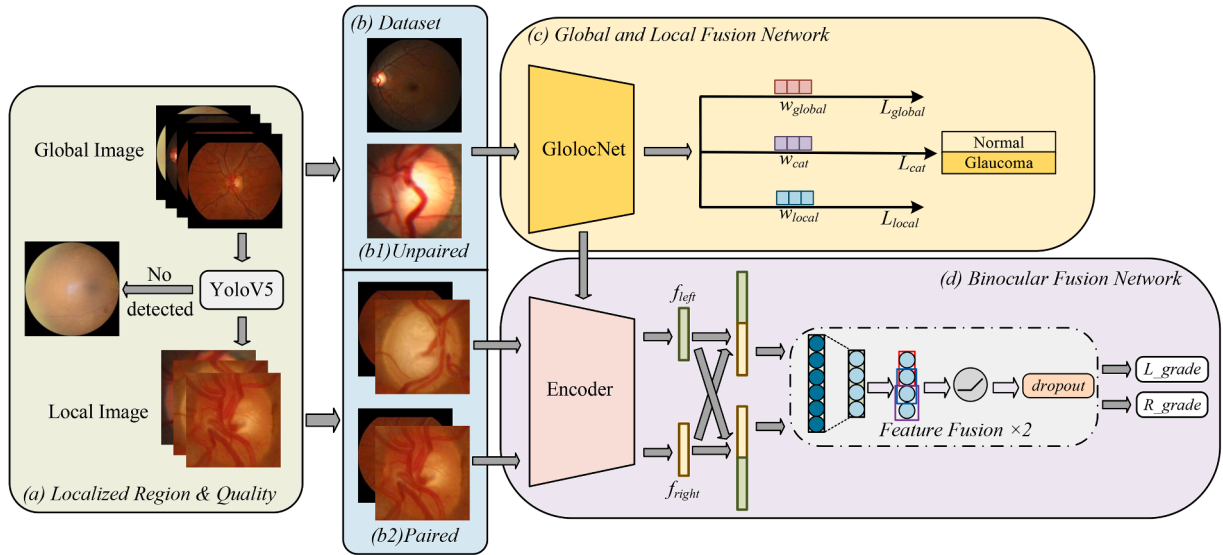


Fig. 2. PK-Net architecture. The overall architecture is primarily composed of (a) Localized region & quality: including the extraction of the local optic disc region from the input global fundus image and quality screening based on YOLOv5; (b) Dataset: consisting of Unpaired single-eye global and local fundus images and paired binocular global and local fundus images; (c) Global and local fusion network (GloLocNet) with triple loss function; and (d) Binocular fusion network (BFNet): including the encoder, left and right eye feature concatenation, and left and right eye feature fusion modules.

Our contributions are summarized as follows:

- **We propose PK-Net to enhance glaucoma screening:** PK-Net is an integrated screening model from data screening to classification. It uses prior knowledge of glaucoma such as the importance of the optic disc and binocular relationship to drive model design to enhance glaucoma screening.
- **We developed GloLocNet to strengthen the feature extraction capabilities:** By combining high-resolution local optic disc images with global fundus images, GloLocNet utilizes a triple-loss optimization strategy to enhance the extraction of features from different pathways.
- **We introduce BFNet for binocular feature fusion:** By exploiting the strong correlation between left and right eye images, BFNet further improves the sensitivity and generalization of glaucoma screening across diverse datasets.
- **We achieve state-of-the-art results on multiple datasets:** It outperforms existing methods by achieving the highest AUC, balanced accuracy (BAcc), and sensitivity (Sen) on three glaucoma screening datasets, demonstrating its potential to advance glaucoma detection.

3. Method

3.1. Overview of PK-net

In this study, we propose a prior knowledge-driven dual-path network (PK-Net) for enhanced glaucoma screening, as illustrated in Fig. 2. Given a global fundus image x_{global} , a high-resolution local optic disc region x_{local} is extracted using a square-cropping strategy detailed in Algorithm 1. To ensure input consistency and maintain data quality, images in which the optic disc cannot be successfully detected are excluded from subsequent analysis.

Leveraging the diagnostic importance of the optic disc region, we introduce the Global and Local Fusion Network (GloLocNet). For unpaired monocular fundus images, GloLocNet processes both x_{global} and x_{local} through two parallel feature extractors to obtain representations f_{global} and f_{local} . These features are adaptively fused and optimized using a triple-loss strategy, which ensures that each path, including the global, local, and concatenated representations, contributes effectively to the final prediction.

To further exploit the pathological correlation between eyes, we design the Binocular Fusion Network (BFNet). For paired-eye samples, BFNet employs two shared-weight GloLocNet encoders to process $(x_{global}^L, x_{local}^L)$ and $(x_{global}^R, x_{local}^R)$, yielding feature embeddings f_{left} and f_{right} . These are fused through a symmetric fusion strategy to generate joint classification outputs for both eyes: L_{grade} and R_{grade} . This architecture allows the model to capture inter-eye dependencies, which are clinically relevant in glaucoma diagnosis.

In summary, PK-Net incorporates domain-specific prior knowledge through dual-path processing and inter-eye fusion, enabling robust, generalizable screening performance for both unpaired and paired fundus image scenarios.

Algorithm 1 Optic disc boundary extraction and cropping for glaucoma screening.

PROCESS:

1. Select the first bounding box $b = (x_1, y_1, x_2, y_2)$.
2. Compute bounding box dimensions:
 $w = x_2 - x_1, \quad h = y_2 - y_1$.
3. Determine square crop dimensions:
 $side_length = \max(w, h)$.
4. Compute square crop center:
 $x_c = \frac{x_1 + x_2}{2}, \quad y_c = \frac{y_1 + y_2}{2}$.
5. Adjust crop boundaries:

$$x'_1 = \max(0, x_c - \frac{side_length}{2}), y'_1 = \max(0, y_c - \frac{side_length}{2}),$$

$$x'_2 = \min(W, x_c + \frac{side_length}{2}), y'_2 = \min(H, y_c + \frac{side_length}{2}),$$

where W, H are the width and height of the original image.

6. Extract the square crop: $C = I[y'_1 : y'_2, x'_1 : x'_2]$.

Return C

3.2. Localized optic disc region extraction and quality screening

To enhance the accuracy of glaucoma screening, we employed a YOLOv5-based algorithm to detect and localize the optic disc region

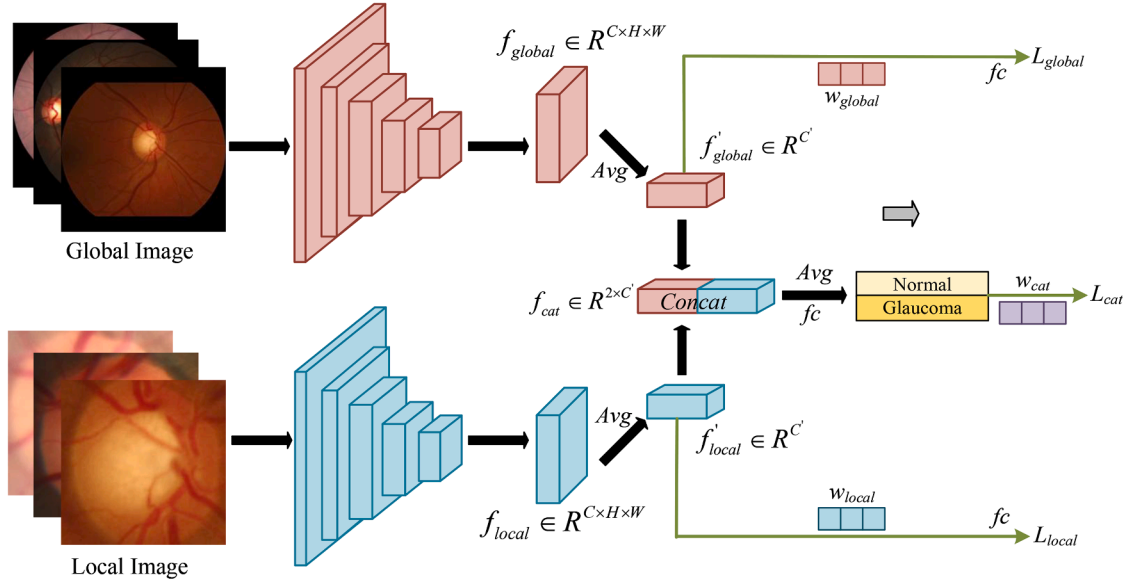


Fig. 3. GloLocNet architecture. (C represents the number of channels, H represents the height, W represents the width, w denotes learnable weight parameters, and Avg stands for Average pooling).

from fundus images. The model was trained on an annotated dataset of fundus images to ensure robust performance in recognizing optic disc regions under varying image conditions. Upon successful identification, the bounding box coordinates provided by YOLOv5 were processed using Algorithm 1 to generate the final cropping boundaries. Specifically, the dimensions of the bounding box were utilized to compute a square region centered on the optic disc, ensuring that the cropped area adhered to spatial constraints required for subsequent analysis.

If the optic disc was not detected, the corresponding image was excluded from further analysis to maintain data quality. This filtering step reduced computational overhead and enhanced the reliability of diagnostic metrics. Algorithm 1 details the procedures for cropping and resizing the optic disc region, standardizing the extracted region to a uniform size. This process mitigates inconsistencies in optic disc region dimensions across different datasets, addressing a critical requirement for accurate glaucoma screening. The specific effect is shown in Fig. 7.

3.3. Global and local fusion network GloLocNet

GloLocNet is a dual-path fusion network designed specifically for glaucoma screening, leveraging both global and local information from fundus images, as shown in Fig. 3. Traditional deep learning methods often focus on either global or local information, but GloLocNet addresses the limitations of these approaches by considering both simultaneously.

The network uses a two-level information fusion strategy: global images provide a broad structural view of the fundus, while local optic disc region images, cropped using YOLOv5, focus on detailed features. GloLocNet can independently process unpaired single-eye images and also serve as an encoder for a binocular fusion network.

In the dual-path network, we design a feature extractor for each path, using pre-trained ConvNeXt [39] models. The feature maps output by each feature extractor are converted into one-dimensional features through an adaptive average pooling layer, followed by a fully connected layer for classification.

Specifically, both the global image path f_{global} and the local image path f_{local} undergo feature extraction through their respective ConvNeXt models:

$$f_{global} = \Phi(\text{ConvNeXt}(x_{global}; \theta_{global})) \quad (1)$$

$$f_{local} = \Phi(\text{ConvNeXt}(x_{local}; \theta_{local})) \quad (2)$$

Where Φ represents the activation function, such as ReLU, and θ_{global} , θ_{local} are the model parameters for each path. The extracted features f_{global} and f_{local} are then processed by adaptive average pooling and converted into one-dimensional vectors:

$$f'_{global} = \text{AdaptiveAvgPool}(f_{global}) \quad (3)$$

$$f'_{local} = \text{AdaptiveAvgPool}(f_{local}) \quad (4)$$

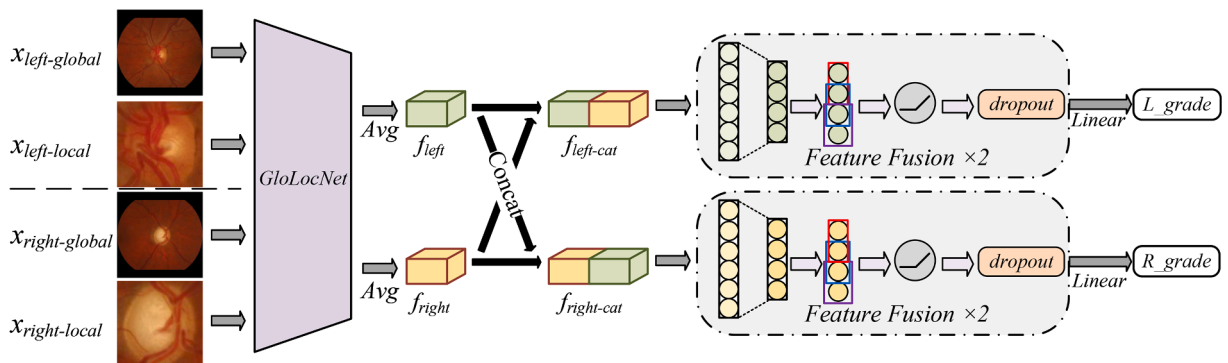


Fig. 4. BFNet Architecture. (L_grade represents the screening result for the left eye, and R_grade represents the screening result for the right eye).

To integrate global and local information, we concatenate the feature vectors f'_{local} and f'_{global} pass them through another fully connected layer to generate the final classification result:

$$f_{cat} = \text{Concat}(f'_{global}, f'_{local}) \quad (5)$$

Lastly, f_{cat} is fed into a fully connected layer to produce the final classification result.

3.4. Binocular fusion network (BFNet)

In this study, we introduce a Binocular Fusion Network (BFNet) based on our previously developed GloLocNet (Global and Local Fusion Network) model, as shown in Fig. 4. This network is specifically designed to process fundus images from both eyes, enhancing the diagnostic accuracy for glaucoma and other retinal diseases by integrating global and local information from both eyes. The BFNet network consists of two modified GloLocNet encoders, each processing the global and local images of one eye. These encoders have been adjusted by removing the pre-avgpool components from the original GloLocNet to facilitate feature fusion. The operation of the BFNet is as follows:

The global and local images of both eyes are processed by their corresponding GloLocNet encoders, independently extracting essential visual features:

$$f_{left} = \text{GloLocNet}(x_{left-global}, x_{left-local}) \quad (6)$$

$$f_{right} = \text{GloLocNet}(x_{right-global}, x_{right-local}) \quad (7)$$

The features extracted by the encoders are then fused and further processed and classified by a features exaction module.

If the image sequence is (left eye, right eye), the features are concatenated in the order of f_{left} and f_{right} , and the left eye classification result is then obtained using the function $F(x)$.

$$f_{left-cat} = \text{Concat}(f_{left}, f_{right}) \quad (8)$$

$$L_{grade} = F(f_{left-cat}) \quad (9)$$

Conversely, if the image sequence is (right eye, left eye), the features are concatenated in the order of f_{right} and f_{left} , and the right eye classification result is then obtained using the function $F(x)$.

$$f_{right-cat} = \text{Concat}(f_{right}, f_{left}) \quad (10)$$

$$R_{grade} = F(f_{right-cat}) \quad (11)$$

Here, represents the Algorithm 2 for feature fusion and classification, defined as follows:

Algorithm 2 $F(x)$ for feature fusion and classification.

```

PROCESS:
# Perform sequential operations on input feature vector x
x = Linear(x, in_features, num_features//2) #First linear transformation
x = MaxPool1d(x, 2) #Max pooling with kernel size of 2
x = ReLU(x) #Activation function
x = Dropout(x, 0.3) # Dropout with probability 0.3
x = Linear(x, num_features//4, num_features//8) #Second linear transformation
x = MaxPool1d(x, 2) #Max pooling with kernel size of 2
x = ReLU(x) # Activation function
x = Dropout(x, 0.3) # Dropout with probability 0.3
output = Linear(x, num_features//8, output_features) #Final linear transformation to 2 output features
Return output

```

3.5. Loss

To enhance generalization and mitigate overfitting, we adopted Label Smoothing Cross Entropy Loss as the optimization function. This

approach introduces label smoothing into the traditional cross-entropy loss, reducing model overconfidence and improving performance on imbalanced datasets. The smoothed label vector y' is computed as:

$$y' = (1 - \alpha) \cdot y + \frac{\alpha}{C} \quad (12)$$

Where y is the original one-hot encoded label vector, α is the smoothing parameter (a small constant between 0 and 1), and C denotes the total number of classes. This adjustment redistributes a portion of the label's probability mass uniformly across all classes.

The cross-entropy loss with smoothed labels is defined as:

$$Loss = - \sum_{i=1}^C y'_i \log(p_i) \quad (13)$$

Where y'_i is the i th component of the smoothed label vector, and p_i is the predicted probability for class i , obtained from the model's softmax output.

Building upon this, we refined the training of GloLocNet by designing a triple loss function (see Fig. 3). This function optimizes the outputs from the global, local, and concatenated feature paths by computing individual losses and combining them using adaptive weighting, ensuring balanced contributions and improved robustness.

For each feature path, the output probabilities are computed as:

$$o_i = \text{Softmax}(W_i f_i + b_i), (i = 'global', 'local', 'cat') \quad (14)$$

Where f_i is the feature vector from each path, and W_i and b_i are learnable weight and bias parameters.

The total loss combines the individual path losses with weights w_i :

$$Loss = \sum_i (w_i \cdot L_i(o_i, y)), i = \{'global', 'local', 'cat'\} \quad (15)$$

Where L_i represents the label-smoothed cross-entropy loss for each path's output o_i , and y is the true label. The weights w_i are trainable parameters that balance the contributions of each path to the overall loss.

This integrated loss function leverages the strengths of individual feature paths and enhances the model's ability to learn complementary information from global and local contexts. Incorporating label smoothing further improves robustness to noise and overfitting, leading to better performance on unseen data.

4. Experiments

4.1. Datasets

Our experimental analysis utilized three distinct datasets: the paired binocular PAPILA dataset, the ODIR dataset, and a combined dataset comprising the ORIGA dataset for training and the REFUGE dataset for testing. The details of these datasets are shown in Table 1.

PAPILA dataset [40] contains fundus images and clinical data from 244 patients, with images from both eyes of each patient. This dataset is specifically designed to support research on early glaucoma diagnosis by leveraging comprehensive data from both eyes. Additionally, it includes segmentation information for each patient's optic disc and cup, alongside diagnostic outcomes based on clinical data. For our analysis, we focused on images labeled as normal (0) and glaucoma (1), selecting data from 210 patients. The subset included 333 non-glaucoma images and 87 glaucoma images. A five-fold cross-validation approach was employed, ensuring that all images from a single patient were consistently assigned to either the training or validation set.

OIA-ODIR [41] dataset is a comprehensive public resource containing over 8000 high-resolution fundus images from more than 5000 patients. This dataset includes various ocular diseases such as glaucoma, diabetic retinopathy, and age-related macular degeneration. Alongside retinal images, it provides associated clinical information. For our study, we selected 223 high-quality fundus images of glaucoma patients from the training set, along with 421 images from normal individuals and

Table 1
Detailed description.

Dataset	Characteristic	Category	Quantity	Total
PAPILA [40]	Paired	Normal/Glaucoma	333/87	420
OIA-ODIR [41]	Paired	Normal/Glaucoma	314/586	900
ORIGA [42]	Unpaired	Normal/Glaucoma	482/168	650
REFUGE [43]	Unpaired	Normal/Glaucoma	720/80	800
GAMMA [44]	Unpaired	Normal/Early stage/Mid-advanced stage	50/26/24	100

patients with other ocular diseases. From the test set, we selected 91 quality-screened fundus images of glaucoma patients and 165 images from normal individuals and those with other ocular diseases. Notably, all glaucoma patients in both the training and test sets were diagnosed with glaucoma in both eyes.

ORIGA dataset [42] is a specialized resource for glaucoma analysis, containing 650 high-resolution retinal fundus images, including 168 glaucoma and 482 normal images. Curated by the Singapore National Eye Centre (SNEC) and the National University of Singapore (NUS), this dataset provides detailed diagnostic labels and related clinical information, serving as a valuable resource for training glaucoma detection models.

REFUGE dataset [43] is specifically curated for glaucoma detection and analysis, featuring 1200 high-resolution retinal fundus images annotated with optic disc and cup regions. This dataset, collected by various international research institutions and hospitals, is aimed at advancing automated glaucoma detection and image segmentation. For our analysis, we used fundus images from the training and validation sets as our test set, which included 720 normal and 80 glaucoma images, as the test set labels were not provided.

GAMMA dataset [44] is the first publicly available multi-modal glaucoma grading dataset, provided by the Sun Yat-sen Ophthalmic Center, Sun Yat-sen University. It includes data from 300 patients, comprising paired 2D color fundus photographs and 3D optical coherence

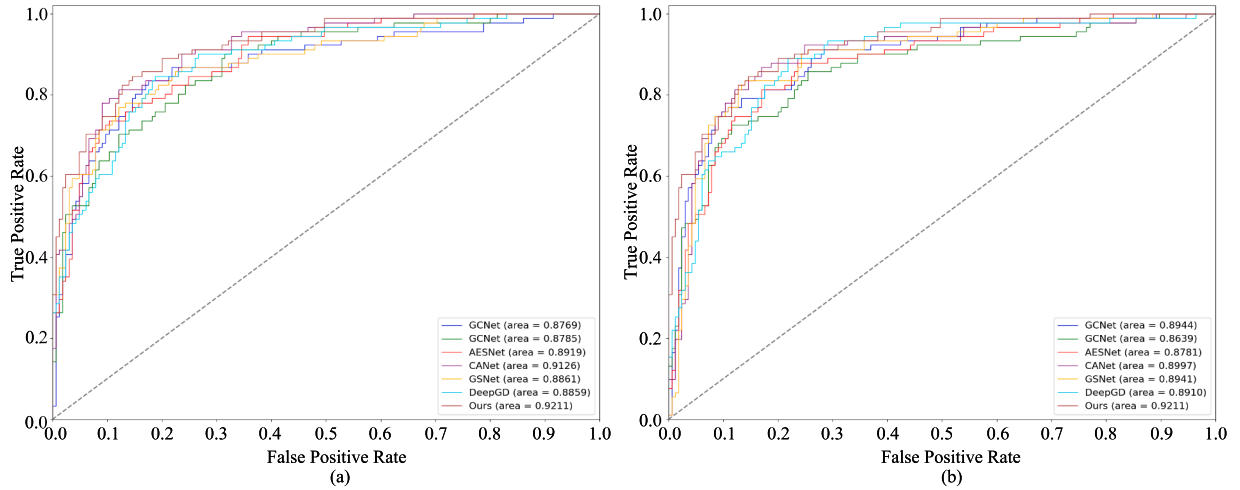


Fig. 5. ROC curves on the OIA-ODIR dataset: (a) Local image, (b) Global image.

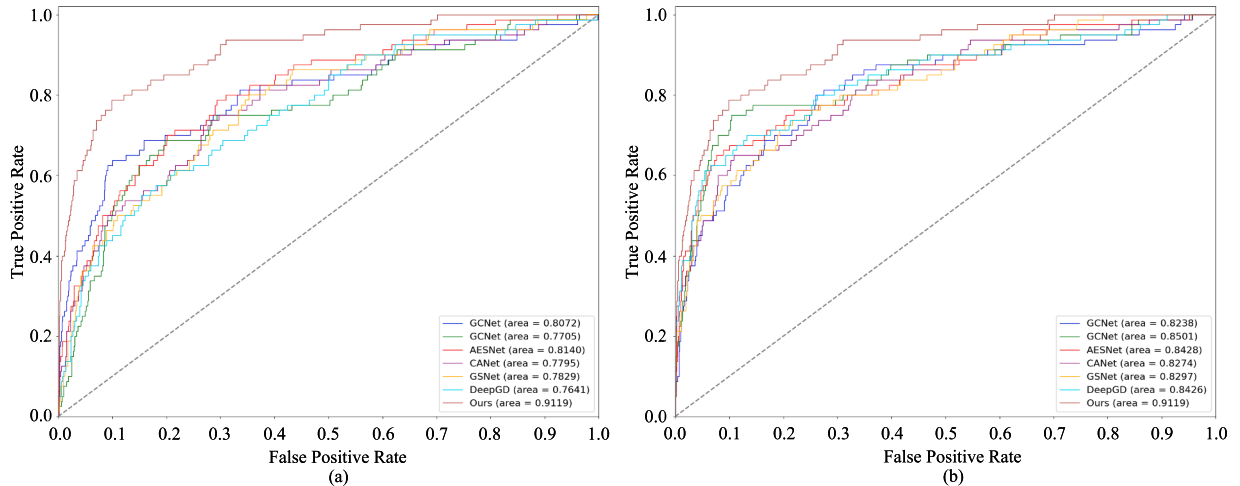


Fig. 6. ROC curves trained on the ORIGA dataset and tested on the REFUGE dataset: (a) Local image, (b) Global image.

Table 2

Comparisons with state-of-the-art methods on the PAPILA dataset. (Local represents the input as the local optic disc region image, and Global represents the input as the global image).

Method	Local					Global				
	AUC	BAcc	Sen	Spe	p-value	AUC	BAcc	Sen	Spe	p-value
CAnet [28]	0.8054	0.666	0.4022	0.9297	<0.05	0.9004	0.7737	0.5995	0.9480	<0.05
GCnet(resnet50) [29]	0.8066	0.7097	0.5048	0.9147	<0.05	0.9034	0.8021	0.6640	0.9402	<0.05
GCnet(InceptionV3) [29]	0.8683	0.7359	0.5196	0.9521	<0.05	0.8670	0.7484	0.6035	0.8933	<0.05
AESNet [31]	0.7726	0.6505	0.3779	0.9230	<0.05	0.9002	0.7528	0.5447	0.9609	<0.05
GSNet [30]	0.8803	0.7717	0.6089	0.9345	<0.05	0.9106	0.7966	0.6530	0.9609	<0.05
DeepGD [32]	0.8291	0.7147	0.5091	0.9203	<0.05	0.8715	0.7817	0.6448	0.9185	<0.05
BFNet (Ours)	0.9567	0.9118	0.8872	0.9363	–	0.9567	0.9118	0.8872	0.9363	–

Table 3

Comparisons with state-of-the-art methods on the OIA-ODIR dataset.

Method	Local					Global				
	AUC	BAcc	Sen	Spe	p-value	AUC	BAcc	Sen	Spe	p-value
CAnet [28]	0.9028	0.8380	0.7912	0.8848	<0.05	0.8997	0.8356	0.7802	0.8909	<0.05
GCnet(resnet50) [29]	0.8769	0.7922	0.6813	0.9030	<0.05	0.8944	0.8229	0.7912	0.8545	<0.05
GCnet(InceptionV3) [29]	0.8785	0.7850	0.7033	0.8667	<0.05	0.8639	0.7996	0.7143	0.8848	<0.05
AESNet [31]	0.8919	0.8015	0.7363	0.8667	<0.05	0.8781	0.8100	0.7473	0.8727	<0.05
GSNet [30]	0.8861	0.8179	0.7692	0.8667	<0.05	0.8941	0.8049	0.8462	0.7636	<0.05
DeepGD [32]	0.8859	0.7746	0.6703	0.8788	<0.05	0.8910	0.7869	0.7253	0.8485	<0.05
BFNet (Ours)	0.9211	0.8468	0.8571	0.8364	–	0.9211	0.8468	0.8571	0.8364	–

Table 4

Comparisons of state-of-the-art methods trained on the ORIGA dataset and tested on the REFUGE dataset.

Method	Local					Global				
	AUC	BAcc	Sen	Spe	p-value	AUC	BAcc	Sen	Spe	p-value
CAnet [28]	0.7795	0.7028	0.6125	0.7931	<0.05	0.8274	0.7562	0.6000	0.9125	<0.05
GCnet(resnet50) [29]	0.7706	0.7229	0.5750	0.8708	<0.05	0.8238	0.7542	0.6875	0.8208	<0.05
GCnet(InceptionV3) [29]	0.8072	0.6722	0.3750	0.9694	<0.05	0.8501	0.8063	0.7125	0.9000	<0.05
AESNet [31]	0.8140	0.7194	0.5625	0.8764	<0.05	0.8428	0.7646	0.5875	0.9417	<0.05
GSNet [30]	0.7891	0.6861	0.5750	0.7972	<0.05	0.8297	0.7243	0.5000	0.9486	<0.05
DeepGD [32]	0.7641	0.6979	0.6125	0.7833	<0.05	0.8426	0.7799	0.6250	0.9347	<0.05
GloLocNet (Ours)	0.9119	0.8382	0.7875	0.8889	–	0.9119	0.8382	0.7875	0.8889	–

Table 5

Comparisons with state-of-the-art methods on the GAMMA dataset.

Method	Local					Global				
	AUC	BAcc	Sen	Spe	p-value	AUC	BAcc	Sen	Spe	p-value
CAnet [28]	0.9047	0.7501	0.6467	0.8536	<0.05	0.8975	0.7701	0.6755	0.8647	<0.05
GCnet(resnet50) [29]	0.9138	0.7386	0.6267	0.8505	<0.05	0.8965	0.6558	0.5200	0.7917	<0.05
GCnet(InceptionV3) [29]	0.8914	0.7193	0.6044	0.8342	<0.05	0.9139	0.6811	0.5411	0.8211	<0.05
AESNet [31]	0.9110	0.7619	0.6622	0.8615	<0.05	0.9019	0.7722	0.6767	0.8678	<0.05
GSNet [30]	0.9130	0.8099	0.7244	0.8955	<0.05	0.9059	0.7440	0.6344	0.8535	<0.05
DeepGD [32]	0.8910	0.7719	0.6689	0.8749	<0.05	0.8895	0.7533	0.6467	0.8599	<0.05
GloLocNet (Ours)	0.9516	0.8337	0.7549	0.9129	–	0.9516	0.8337	0.7549	0.9129	–

Table 6

The 5-Fold cross-validation result of GloLocNet and BFNet on the PAPILA dataset.

Model	AUC	BAcc	Sen	Spe
GloLocNet	0.9169	0.8367	0.7372	0.9361
BFNet	0.9567	0.9118	0.8872	0.9363

Table 7

Ablation study of model inputs on the paired binocular OIA-ODIR dataset.

Input	AUC	BAcc	Sen	Spe
Global	0.8896	0.8278	0.8132	0.8424
Local	0.9056	0.8322	0.8462	0.8182
GloLocNet	0.9100	0.8339	0.8132	0.8545
BFNet	0.9211	0.8468	0.8571	0.8364

Table 8

Ablation study of input types on unpaired monocular data: trained on ORIGA and tested on REFUGE.

Input	AUC	BAcc	Sen	Spe
Global	0.8756	0.7875	0.8500	0.7250
Local	0.8549	0.7465	0.6750	0.8181
GloLocNet (Single loss)	0.9080	0.8472	0.8125	0.8819
GloLocNet (Triple loss)	0.9172	0.8507	0.7625	0.9389

tomography (OCT) scans. Annotations include glaucoma grading labels, macular center coordinates in the fundus images, and optic disc/cup segmentation masks. In this study, we utilized 100 color fundus images from Task 1 of the training set, categorized into three classes: non-glaucoma, early glaucoma, and mid-to-advanced glaucoma.

Table 9

Comparisons of experimental results for state-of-art models using TLD and OLD extraction methods on the unpaired single eye dataset. (TLD:their local optic disc extraction method, OLD: our local optic disc extraction method).

Method	TLD				OLD			
	AUC	BAcc	Sen	Spe	AUC	BAcc	Sen	Spe
CAnet [28]	0.6663	0.6243	0.6000	0.6486	0.7795	0.7028	0.6125	0.7931
GCnet(resnet50) [29]	0.6633	0.6194	0.6625	0.5764	0.7706	0.7229	0.5750	0.8708
GCnet(InceptionV3) [29]	0.7095	0.6611	0.6750	0.6472	0.8072	0.6722	0.3750	0.9694
AESNet [31]	0.7594	0.6424	0.8125	0.4722	0.8140	0.7194	0.5625	0.8764
GSNet [30]	0.6766	0.5667	0.7750	0.3583	0.7891	0.6861	0.5750	0.7278
DeepGD [32]	0.7176	0.6535	0.7250	0.5819	0.7641	0.6979	0.6125	0.7833
GloLocNet(Ours)	0.8983	0.8188	0.7125	0.9250	0.9119	0.8382	0.7875	0.8889

Table 10

Comparisons of experimental results for different base models as backbones for GloLocNet and BFNet using TLD and OLD extraction methods on unpaired single-eye datasets.

Backbone	GloLocNet (TLD)				GloLocNet (OLD)			
	AUC	BAcc	Sen	Spe	AUC	BAcc	Sen	Spe
EfficientNetB0 [47]	0.8357	0.7361	0.5625	0.9097	0.8590	0.7889	0.7000	0.8778
DenseNet121 [48]	0.8622	0.7979	0.6750	0.9208	0.9062	0.8083	0.8125	0.8042
ResNet50 [49]	0.8754	0.8028	0.7375	0.8681	0.8996	0.8354	0.7750	0.8958
ConvNeXt_Tiny [39]	0.8983	0.8188	0.7125	0.9250	0.9119	0.8382	0.7875	0.8889

Table 11

Performance comparisons of dual-input extended baseline methods and BFNet on the PAPILA dataset.

Method	Params	FLOPs	Train time	Test time	AUC	BAcc	Sen	Spe
CAnet [28]	24.5M	30.333G	13.13s	0.0242s	0.9017	0.8176	0.7217	0.9081
GCnet(resnet50) [29]	67.95M	43.198G	9.39s	0.0243s	0.9186	0.8213	0.7363	0.9063
GCnet(InceptionV3) [29]	71.16M	35.283G	10s	0.0205s	0.9035	0.7969	0.6740	0.9199
AESNet [31]	51.98M	39.094G	17.49s	0.0281s	0.8954	0.8248	0.8120	0.8377
GSNet [30]	22.79M	30.938G	13.28s	0.0276s	0.9166	0.8100	0.7101	0.9099
DeepGD [32]	38.69M	16.45G	12.55s	0.020s	0.8903	0.7695	0.6065	0.9333
BFNet (Ours)	62.2M	93.280G	44.34s	0.0279s	0.9567	0.9118	0.8872	0.9363

Table 12

Cross-dataset performance of dual-input extended baseline methods and GloLocNet (trained on ORIGA, tested on REFUGE).

Method	Params	Flops	Train time	Test time	AUC	BAcc	Sen	Spe
CAnet [28]	24.5M	30.333G	21.20s	0.0082s	0.8747	0.7924	0.7375	0.8472
GCnet(resnet50) [29]	67.95M	43.198G	15.31s	0.0078s	0.8429	0.7882	0.7500	0.8264
GCnet(InceptionV3) [29]	71.16M	35.283G	16.44s	0.0071s	0.8617	0.7854	0.7875	0.7833
AESNet [31]	51.98M	39.094G	28.43s	0.0090s	0.8761	0.8042	0.7125	0.8958
GSNet [30]	22.79M	30.938G	21.34s	0.0083s	0.8605	0.7674	0.6750	0.8597
DeepGD [32]	38.69M	16.45G	19.33s	0.0067s	0.8623	0.7889	0.7125	0.8653
GloLocNet (Ours)	57.18M	46.637G	22.19s	0.0073s	0.9119	0.8382	0.7875	0.8889

Table 13

Cross-dataset comparisons under a fixed sensitivity of 0.95: dual-input baseline methods vs. GloLocNet (trained on ORIGA, tested on REFUGE).

Method	BAcc	Spe
CAnet [28]	0.6757	0.4014
GCnet(resnet50) [29]	0.6132	0.2764
GCnet(InceptionV3) [29]	0.6597	0.3694
AESNet [31]	0.6160	0.2819
GSNet [30]	0.6493	0.3486
DeepGD [32]	0.6792	0.4083
GloLocNet(Ours)	0.7285	0.5069

Table 14

Performance comparisons under a fixed sensitivity of 0.95: dual-input baseline methods vs. BFNet on OIA-ODIR dataset.

Method	BAcc	Spe
CAnet [28]	0.7144	0.4788
GCnet(resnet50) [29]	0.6568	0.3636
GCnet(InceptionV3) [29]	0.6083	0.2667
AESNet [31]	0.7083	0.4667
GSNet [30]	0.7508	0.5515
DeepGD [32]	0.7023	0.4545
BFNet(ours)	0.7841	0.6182

4.2. Implementation details

Our model framework was pre-trained on the ImageNet-1k dataset [45]. Initially, we applied the Hough Circle Transform to crop the Field of View (FoV) and resized the images by padding zeros to the shorter

edge, ensuring balanced width and height in the cropped regions. The images were then resized to 512×512 pixels. To mitigate overfitting, we incorporated several data augmentation techniques, including random rotation, horizontal flipping, and vertical flipping.

For network optimization, we began training with an initial learning rate of 0.0001, conducting the training over 50 epochs. We employed the ReduceLROnPlateau scheduler to dynamically adjust the learning rate, enhancing the training process. The Adam optimizer [46] was used in conjunction with a multi-loss cross-entropy loss function, and we set the batch size to 8. All implementations were executed on an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory using the PyTorch framework.

4.3. Evaluation metrics

For evaluation, we report the Receiver Operating Characteristic (ROC) curve and the area under the ROC curve (AUC). In addition, we use three key metrics to assess model performance: sensitivity (Sen), specificity (Spe), and balanced accuracy (BAcc). These metrics are defined as follows:

$$Sen = \frac{TP}{TP + FN}, Spe = \frac{TN}{TN + FP}, BAcc = \frac{Sen + Spe}{2} \quad (16)$$

Here, TP (True Positives) and TN (True Negatives) represent the number of correctly identified positive and negative cases, respectively, while FP (False Positives) and FN (False Negatives) represent the number of incorrect positive and negative classifications. To determine the optimal diagnostic threshold, we adjust it across a range of values to obtain corresponding metric scores. The threshold that produces the highest BAcc score is selected as the final threshold for reporting the model's performance.

For statistical analysis, given our a priori hypothesis that the proposed method would outperform the baseline methods in terms of AUC, and considering the limited sample size in our evaluation, we conducted one-tailed paired tests. The corresponding p-values are reported in the results tables, where $p < 0.05$ is considered statistically significant.

4.4. Comparison with the state-of-the-art methods

In this glaucoma screening task, we evaluated the performance of several models across three different datasets. For the first two datasets, which included paired binocular fundus images, we applied the Binocular Fusion Network (BFNet) method. For the third dataset, which consisted of individual fundus images, we employed the GloLocNet method, optimizing the model through independent loss calculations on each path using a triple loss strategy. It is important to note that the results of other models were reproduced based on their respective papers, where the input was the localized optic disc region. The results of these experiments are presented in the "Local" columns of Tables 2–5 as well as in Figs. 5 and 6.

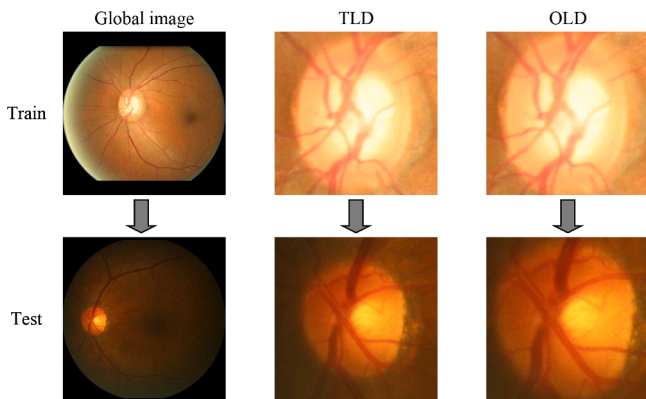


Fig. 7. Comparison of extracted images using their local optic disc extraction method (TLD) and our local optic disc extraction method (OLD) (train represents the ORIGA dataset, test represents the REFUGE dataset).

Our method demonstrated significant advantages across all three datasets, particularly in terms of AUC, BAcc, and sensitivity. On the first two datasets, which included paired binocular images, the BFNet method significantly enhanced detection performance. On the third dataset, which did not include paired images, the GloLocNet method combined with the triple loss strategy maintained high recognition accuracy. These results indicate that our approach effectively improves the accuracy and reliability of glaucoma screening in both paired and single-eye scenarios.

Additionally, we conducted experiments where other models were compared using 512×512 global images as input. The results of these experiments are shown in the "Global" columns of Tables 2–5. Our model continued to demonstrate significant performance advantages across all four datasets, particularly in AUC, BAcc, and sensitivity. Whether using paired binocular datasets or single-eye datasets, our model, through the BFNet or GloLocNet method combined with the triple loss strategy, consistently improved the accuracy and robustness of glaucoma screening.

4.5. Ablation studies

- (1) Ablation and comparative analysis on paired binocular datasets:** Tables 6 and 7 present the results of the proposed GloLocNet and the binocular fusion network (BFNet) across two paired binocular datasets. On the PAPILA dataset, BFNet demonstrates superior overall performance compared to GloLocNet, achieving an AUC of 0.9567 and a balanced accuracy (BAcc) of 0.9118, with notable gains in sensitivity (0.8872 vs. 0.7372) while maintaining a comparable specificity (0.9363 vs. 0.9361). This suggests that incorporating binocular information effectively enhances the model's capability to identify glaucoma-positive cases without sacrificing its ability to correctly classify negative cases.

The ablation study on the paired binocular OIA-ODIR dataset further validates the benefits of integrating global and local information as well as binocular fusion. When only the global view is used, the model attains an AUC of 0.8896, while using only the local view yields a slightly higher AUC of 0.9056, indicating the diagnostic relevance of the optic disc region. The proposed GloLocNet, which combines both views, achieves an AUC of 0.9100, showing modest improvements in balanced accuracy and specificity compared to single-view inputs. Notably, BFNet outperforms all other configurations, achieving the highest AUC (0.9211) and BAcc (0.8468), with balanced gains in both sensitivity and specificity. These results confirm that dual-path processing, complemented by inter-eye fusion, is crucial for maximizing the screening performance of glaucoma detection systems.

- (2) Ablation analysis on unpaired monocular datasets:** Table 8 reports the ablation study on unpaired monocular data, where models were trained on the ORIGA dataset and tested on the REFUGE dataset to evaluate cross-dataset generalization. Using only the global view yields an AUC of 0.8756 and a balanced accuracy (BAcc) of 0.7875, with relatively high sensitivity (0.8500) but lower specificity (0.7250). In contrast, the local view achieves slightly lower AUC (0.8549) and BAcc (0.7465), but offers improved specificity (0.8181), highlighting its ability to better discriminate negative cases.

The integration of global and local views in GloLocNet substantially improves performance over single-view inputs. Specifically, GloLocNet with a single-loss optimization achieves an AUC of 0.9080 and a BAcc of 0.8472, indicating that combined spatial information enhances both sensitivity and specificity. Further adopting the triple-loss strategy yields the best results, with an AUC of 0.9172 and the highest specificity (0.9389), suggesting that multi-objective optimization effectively strengthens the discriminative capability of the fused representation while preserving generalization across datasets.

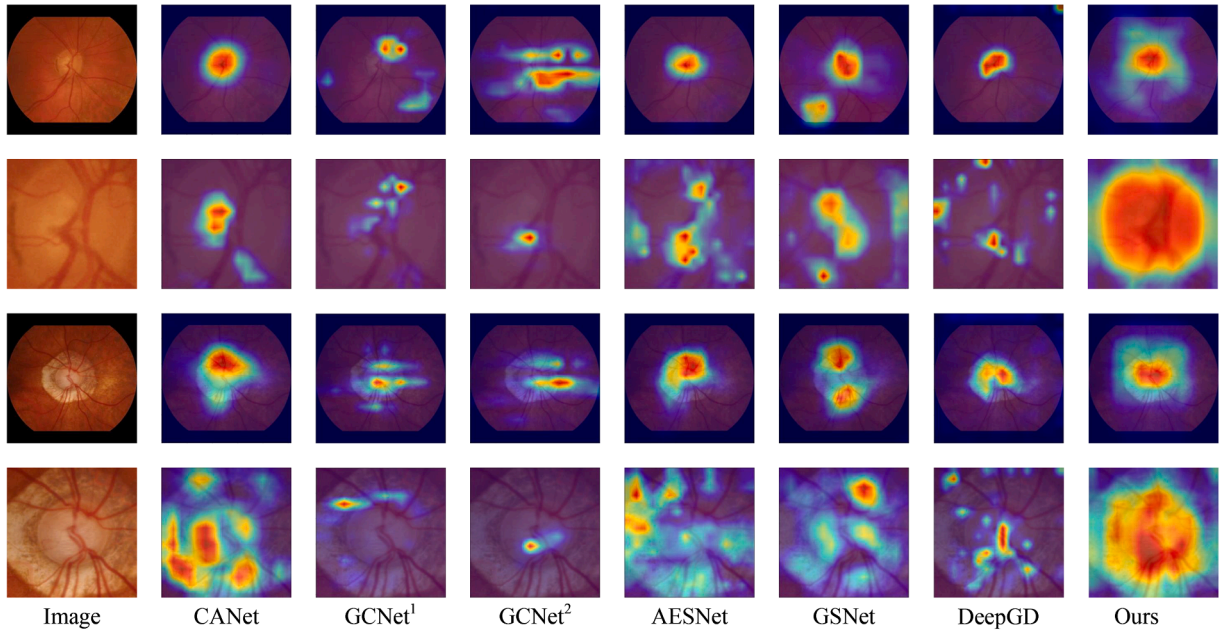


Fig. 8. Qualitative comparison of attention heatmaps generated by CANet, GCNet (ResNet50), GCNet (InceptionV3), AESNet, GSNet, DeepGD, and our proposed method. The first and third rows represent global fundus images, while the second and fourth rows show the corresponding local optic disc images.

(3) Cross-data test results of state-of-the-art models and our BFNet using TLD and OLD optic disc local image cropping strategies:

Fig. 7 illustrates the visual differences between the traditional local optic disc extraction method (TLD) and our proposed optic disc localization and cropping method (OLD), when trained on the ORIGA dataset and tested on the REFUGE dataset. The TLD method, adopted in prior studies, locates the center of the optic disc and applies a fixed 512×512 cropping window. However, this approach does not account for variations in optic disc size, shape, and surrounding background across datasets. As shown in the “test” examples of Fig. 7, this fixed-box strategy frequently includes excessive irrelevant background or truncates important disc structures in cross-dataset scenarios, which can degrade model generalization.

In contrast, our OLD method employs a YOLOv5-based detector to locate the optic cup boundary and generates a square crop based on the longer side of the detected region, which is then resized to 512×512. This design ensures that the extracted local regions preserve consistent structural information of the optic disc across datasets, while minimizing background interference. Such consistency enables models to adapt more effectively to dataset-specific characteristics and improves robustness in cross-dataset testing.

The quantitative results in Table 9 confirm these visual observations. Across all state-of-the-art baselines, replacing TLD with OLD leads to notable gains in AUC, BAcc, and especially specificity. For example, GCNet (InceptionV3) improves specificity from 0.6472 to 0.9694, and AESNet from 0.4722 to 0.8764. These substantial improvements demonstrate that precise, edge-based cropping significantly reduces false positives by focusing on relevant pathological regions. Our GloLocNet achieves the highest overall performance with OLD, reaching an AUC of 0.9119 and a BAcc of 0.8382, outperforming all competing methods under both extraction strategies.

Table 10 further evaluates different backbone architectures within GloLocNet under the two cropping strategies. OLD consistently improves or maintains performance across all backbones, with ConvNeXt_Tiny delivering the best results (AUC = 0.9119, BAcc =

0.8382). Notably, even when using TLD, the dual-path design of GloLocNet mitigates extraction errors by leveraging complementary global image information, resulting in more stable performance compared to single-path baselines.

Overall, the results demonstrate that our OLD method, by directly extracting the optic disc region according to the optic cup boundary and adapting crop dimensions to each image, offers a clear advantage over fixed-box approaches. This strategy not only excels in key metrics such as AUC and specificity but also substantially enhances cross-dataset generalization for glaucoma screening models.

- (4) **Comparative evaluation under consistent Dual-Input configuration and complexity analysis on paired binocular dataset:** To ensure a consistent and fair comparison, all baseline methods were re-implemented in a dual-input configuration, enabling them to process both global fundus images and corresponding local optic disc patches simultaneously. This setting allows the models to fully exploit complementary spatial information, aligning with the input strategy of our proposed BFNet. We evaluated model performance in terms of both classification accuracy—measured by AUC, BAcc, sensitivity (Sen), and specificity (Spe)—and computational complexity, including parameter count, FLOPs, average training time per epoch, and average inference time per image. The results on the PAPILA dataset are summarized in Table 11.

Our BFNet achieves the best classification performance among all compared models, attaining an AUC of 0.9567 and a BAcc of 0.9118, with balanced improvements in both Sen (0.8872) and Spe (0.9363). These gains indicate that integrating binocular information with global–local fusion effectively enhances the model’s ability to detect glaucoma-positive cases without sacrificing specificity. Although BFNet incurs a higher computational cost (93.28G FLOPs; 44.34s per training epoch) due to the processing of paired left and right eye inputs—each with global and local views—it maintains a relatively compact parameter size (62.20M) by adopting a shared-weight encoder. This design allows BFNet to capture inter-eye rela-

tional features efficiently, resulting in superior performance while remaining smaller than many baseline models such as GCNet with ResNet50 or InceptionV3 backbones.

- (5) **Comparative evaluation under consistent Dual-Input configuration and complexity analysis on unpaired monocular dataset:** For scenarios where only single-eye images are available without paired left–right correspondence, we evaluated the cross-dataset generalization of GloLocNet and dual-input baseline models by training on the ORIGA dataset and testing on the REFUGE dataset. The results, shown in Table 12, reveal that GloLocNet achieves the highest overall performance, with an AUC of 0.9119 and a BAcc of 0.8382, outperforming all re-implemented baselines. In particular, GloLocNet maintains a balanced trade-off between sensitivity (0.7875) and specificity (0.8889), demonstrating robustness to dataset shifts.

GloLocNet’s superior cross-dataset performance can be attributed to two factors: (1) the fusion of global and local representations, which captures both overall retinal context and fine-grained optic disc details, and (2) the triple-loss optimization strategy, which adaptively balances contributions from the global, local, and fused pathways, enhancing resilience to variations in imaging conditions and dataset characteristics. Furthermore, its moderate computational footprint (46.64G FLOPs, 57.18M parameters) makes it efficient for deployment in single-image screening workflows.

BFNet and GloLocNet are complementary frameworks tailored for different clinical data availability. BFNet excels in paired binocular scenarios by leveraging inter-eye feature relationships, while GloLocNet is optimized for unpaired monocular data with strong cross-dataset generalization. Both achieve competitive or superior accuracy compared to dual-input extended baselines, offering flexibility in deployment while balancing performance and computational demands.

- (6) **High-Sensitivity Evaluation on Unpaired Monocular Datasets (GloLocNet):** To ensure fairness and consistency, all baseline methods were re-implemented in a dual-input configuration, enabling simultaneous processing of both global fundus images and corresponding high-resolution local optic disc patches. This setup matches the input strategy of GloLocNet, ensuring that performance differences arise from model design rather than input disparity.

We conducted cross-dataset evaluation under a fixed sensitivity (Sen) of 0.95, reflecting the high-sensitivity requirement of real-world glaucoma screening, where minimizing false negatives is critical. In this setting, balanced accuracy (BAcc) and specificity (Spe) become key indicators of clinical applicability, as excessive false positives could overburden healthcare workflows. As shown in Table 13, GloLocNet achieves the highest BAcc (0.7285) and Spe (0.5069) among all dual-input baselines, outperforming modified versions of AESNet, CANet, GCNet, GSNet, and DeepGD. This superior performance can be attributed to GloLocNet’s effective fusion of global and local representations, combined with its triple-loss optimization strategy, which encourages better feature alignment and improves decision robustness. Notably, while other baselines exhibit substantial drops in specificity when constrained to high sensitivity, GloLocNet maintains a favorable balance, accurately identifying healthy cases without compromising sensitivity.

- (7) **High-Sensitivity Evaluation on Paired Binocular Datasets (BFNet):** In the paired-eye scenario, we evaluated models on the OIA-ODIR dataset, where each subject provides both left and right eye fundus images. The results in Table 14 demonstrate that BFNet consistently outperforms all dual-input baselines under a fixed sensitivity of 0.95, achieving the highest BAcc (0.7841) and specificity (0.6182). This improvement underscores the advantage of leveraging inter-eye correlations, a clinically relevant prior knowledge in glaucoma diagnosis.

Unlike methods that process each eye independently, BFNet adopts GloLocNet as its backbone and integrates binocular features

through a joint fusion strategy, enabling the model to generate more comprehensive and clinically meaningful predictions. Even under the high-sensitivity constraint, BFNet maintains substantially higher specificity compared to other methods, thereby reducing false positives while preserving sensitivity. This capability is essential for practical glaucoma screening, as it ensures early detection without overburdening clinical workflows with unnecessary follow-up examinations.

- (8) **Qualitative analysis of heat maps generated by different methods:** To further illustrate the effectiveness of our proposed method in feature representation, we provide attention heatmaps in Fig. 8, comparing CANet, GCNet (ResNet50), GCNet (InceptionV3), AESNet, GSNet, and our model. In the global fundus images (Row 1 and Row 3), most models are generally able to highlight the optic disc region, which aligns with glaucoma-relevant anatomical priors. However, in local optic disc images (Row 2), attention-based baseline models tend to focus on small, fragmented regions, often failing to capture the overall disc structure. This limits their ability to learn the cup-to-disc ratio, a crucial feature in glaucoma diagnosis. In contrast, our model accurately highlights the optic nerve head as a whole, providing more spatially consistent and clinically interpretable attention across both global and local views. These results support the conclusion that the proposed architecture, guided by domain knowledge, improves the model’s feature learning capacity in a task-specific manner.

5. Conclusion and discussion

In this study, we propose the Prior Knowledge-Driven Multi-Feature Fusion Network (PK-Net), designed to enhance the performance of generic models for glaucoma screening and improve their generalization across diverse datasets by incorporating domain-specific medical knowledge. PK-Net integrates several innovative components, including the GloLocNet model, which optimizes the extraction of local optic disc regions by combining high-resolution local images with global fundus images. It employs a triple-loss optimization strategy to improve classification robustness. Additionally, the BFNet model leverages the complementary relationship between paired eye images to enhance screening accuracy further. Extensive evaluation across three distinct datasets demonstrates that PK-Net achieves state-of-the-art performance in terms of AUC, balanced accuracy (BAcc), and sensitivity (Sen), validating its effectiveness as a tool for glaucoma detection.

Looking forward, our research will focus on two primary directions. First, we plan to enhance PK-Net’s capabilities by integrating multi-modal data, including intraocular pressure measurements and other clinical indicators, alongside retinal images. This integration will provide a more comprehensive assessment of ocular health, improving the model’s diagnostic capability in complex clinical settings. By leveraging a broader range of diagnostic information, we expect this multimodal approach to further enhance PK-Net’s accuracy and clinical applicability. Second, we aim to improve the model’s generalization ability across large-scale, diverse datasets, addressing challenges posed by variations in feature characteristics, such as differences in optic disc size and color across datasets. We also plan to refine PK-Net to enhance its robustness and stability in real-world clinical applications, ensuring its continued effectiveness despite the inherent variability and complexity of clinical data.

In practical clinical scenarios, the availability of fundus images varies among patients, necessitating a flexible and adaptive screening approach. To address this, our method dynamically selects the appropriate model based on the number of available fundus images. If only a single-eye fundus image is provided, GloLocNet processes the image independently for glaucoma screening. If binocular fundus images are available, BFNet jointly analyzes both images, leveraging inter-eye correlations to enhance diagnostic accuracy. For real-world deployment, the proposed algorithm can be implemented on a server-based system using

Docker, enabling seamless integration into clinical workflows. Physicians can interact with the system through a user-friendly interface, selecting GloLocNet for monocular screening or BFNNet for binocular screening based on the available data. This adaptive framework ensures the system's applicability across diverse clinical settings, offering a scalable and efficient solution for automated glaucoma screening while optimizing medical resource utilization. To enhance the interpretability of the model, we employed Grad-CAM to visualize the attention regions involved in the prediction process. This technique generates heatmaps that help assess whether the model's decision-making aligns with clinically relevant glaucomatous features, such as optic disc cupping. This interpretability mechanism not only increases model transparency but also builds clinical trust in the system's predictions, thereby improving its feasibility for real-world glaucoma screening applications.

Despite its promising performance, PK-Net still presents several limitations that merit further exploration. First, the dual-path architecture significantly increases computational overhead, including FLOPs and training time, which may hinder its applicability in real-time or resource-constrained clinical environments. Second, the current framework exclusively utilizes fundus images, thereby missing the opportunity to enhance diagnostic performance through the integration of multimodal clinical data, such as intraocular pressure measurements, visual field tests, or electronic health records [50,51]. Third, the model operates on static images and lacks temporal modeling capabilities, which are crucial for monitoring disease progression over time [52]. Future work will focus on optimizing the computational efficiency of the model, incorporating multimodal clinical data, and extending the framework to include longitudinal analysis, thereby improving its practicality and clinical utility.

CRedit authorship contribution statement

Xiaoyan Kui: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization; **Zeru Hai:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization; **Beiji Zou:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization; **Yang Li:** Writing – review & editing, Writing – original draft, Validation, Supervision, Funding acquisition, Conceptualization; **Wei Liang:** Supervision, Methodology, Investigation, Conceptualization; **Zuheng Ming:** Validation, Supervision, Conceptualization; **Liming Chen:** Validation, Supervision, Conceptualization.

Data availability

I have shared the link to my data/code at the Attach File step

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the [National Natural Science Foundation of China](#) (No. U22A2034, 62177047), High Caliber Foreign Experts Introduction Plan funded by MOST, Key Research and Development Programs of [Department of Science and Technology of Hunan Province](#) (No. 2024JK2135), Major Program from Xiangjiang Laboratory (No. 23XJ02005), the Scientific Research Fund of [Hunan Provincial Education Department](#) (No.24A0018), Central South University Research Programme of Advanced Interdisciplinary Studies (No. 2023QYJC020), the National Natural Science Foundation Youth Fund

(62402180), the Hunan Provincial Natural Science Foundation Youth Fund (2024JJ6338), and the Changsha Natural Science Foundation (kq2402173).

References

- [1] H.A. Quigley, A.T. Broman, The number of people with glaucoma worldwide in 2010 and 2020, *Br. J. Ophthalmol.* 90 (3) (2006) 262–267.
- [2] N. Zhang, J. Wang, Y. Li, B. Jiang, Prevalence of primary open angle glaucoma in the last 20 years: a meta-analysis and systematic review, *Sci. Rep.* 11 (1) (2021) 13762.
- [3] J. Ge, Recent advances of clinical and basic studies in glaucoma in China in the last five years, *Chin. J. Ophthalmol.* 41 (8) (2005) 710–716.
- [4] J.M. Ahn, S. Kim, K.-S. Ahn, S.-H. Cho, K.B. Lee, U.S. Kim, A deep learning model for the detection of both advanced and early glaucoma using fundus photography, *PLoS ONE* 13 (11) (2018) e0207982.
- [5] J. Cheng, J. Liu, Y. Xu, F. Yin, D.W.K. Wong, N.-M. Tan, D. Tao, C.-Y. Cheng, T. Aung, T.Y. Wong, Superpixel classification based optic disc and optic cup segmentation for glaucoma screening, *IEEE Trans. Med. Imaging* 32 (6) (2013) 1019–1032.
- [6] R. Hemelings, B. Elen, J. Barbosa-Breda, M.B. Blaschko, P. De Boever, I. Stalmans, Deep learning on fundus images detects glaucoma beyond the optic disc, *Sci. Rep.* 11 (1) (2021) 20313.
- [7] E. Pekel Özmen, T. Özcan, Diagnosis of diabetes mellitus using artificial neural network and classification and regression tree optimized with genetic algorithm, *J. Forecast.* 39 (4) (2020) 661–670.
- [8] M.H. Wang, L. Xing, Y. Pan, F. Gu, J. Fang, X. Yu, C.P. Pang, K.K.-L. Chong, C.Y.-L. Cheung, X. Liao, et al., AI-based advanced approaches and dry eye disease detection based on multi-source evidence: cases, applications, issues, and future directions, *Big Data Min. Anal.* 7 (2) (2024) 445–484.
- [9] M.V. Cicinelli, S. Marmamula, R.C. Khanna, Comprehensive eye care-issues, challenges, and way forward, *Indian J. Ophthalmol.* 68 (2) (2020) 316–323.
- [10] S.H. Wong, J.C. Tsai, Telehealth and screening strategies in the diagnosis and management of glaucoma, *J. Clin. Med.* 10 (16) (2021) 3452.
- [11] Z. Wang, P.A. Keane, M. Chiang, C.Y. Cheung, T.Y. Wong, D.S.W. Ting, Artificial intelligence and deep learning in ophthalmology, in: *Artificial Intelligence in Medicine*, Springer, 2022, pp. 1519–1552.
- [12] H. Fu, J. Cheng, Y. Xu, D.W.K. Wong, J. Liu, X. Cao, Joint optic disc and cup segmentation based on multi-label deep network and polar transformation, *IEEE Trans. Med. Imaging* 37 (7) (2018) 1597–1605.
- [13] B. Jin, X. Xu, Wholesale price forecasts of green grams using the neural network, *Asian J. Econ. Bank.* (ahead-of-print) (2024).
- [14] B. Jin, X. Xu, Price forecasting through neural networks for crude oil, heating oil, and natural gas, *Meas. Energy* 1 (1) (2024) 100001.
- [15] B. Jin, X. Xu, Carbon emission allowance price forecasting for China Guangdong carbon emission exchange via the neural network, *Global Finance Rev.* 6 (2024) 3491.
- [16] S. Sekhar, W. Al-Nuaimy, A.K. Nandi, Automated localisation of optic disk and fovea in retinal fundus images, in: 2008 16th European Signal Processing Conference, IEEE, 2008, pp. 1–5.
- [17] J.E.W. Koh, E.Y.K. Ng, S.V. Bhandary, A. Laude, U.R. Acharya, Automated detection of retinal health using PHOG and SURF features extracted from fundus images, *Appl. Intell.* 48 (2018) 1379–1393.
- [18] M.R.K. Mookiah, U.R. Acharya, C.M. Lim, A. Petznick, J.S. Suri, Data mining technique for automated diagnosis of glaucoma using higher order spectra and wavelet energy features, *Knowl. Based Syst.* 33 (2012) 73–82.
- [19] D. Parashar, D.K. Agrawal, Automated classification of glaucoma stages using flexible analytic wavelet transform from retinal fundus images, *IEEE Sens. J.* 20 (21) (2020) 12885–12894.
- [20] S. Maheshwari, R.B. Pachori, U.R. Acharya, Automated diagnosis of glaucoma using empirical wavelet transform and correntropy features extracted from fundus images, *IEEE J. Biomed. Health Inform.* 21 (3) (2016) 803–813.
- [21] B. Jin, X. Xu, Forecasting wholesale prices of yellow corn through the Gaussian process regression, *Neural Comput. Appl.* 36 (15) (2024) 8693–8710.
- [22] B. Jin, X. Xu, Predictions of steel price indices through machine learning for the regional North-East Chinese market, *Neural Comput. Appl.* 36 (33) (2024) 20863–20882.
- [23] B. Jin, X. Xu, Pre-owned housing price index forecasts using Gaussian process regressions, *J. Modell. Manage.* 19 (6) (2024) 1927–1958.
- [24] M. Juneja, S. Thakur, A. Uniyal, A. Wani, N. Thakur, P. Jindal, Deep learning-based classification network for glaucoma in retinal images, *Comput. Electr. Eng.* 101 (2022) 108009.
- [25] K. Kim, J. Ha, GMFLDA: improved prediction of lncRNA-disease association via graph convolutional network, *13 (2025)* 85330–85341.
- [26] J. Ha, Graph convolutional network with neural collaborative filtering for predicting miRNA-disease association, *Biomedicines* 13 (1) (2025) 136.
- [27] H. Fu, J. Cheng, Y. Xu, C. Zhang, D.W.K. Wong, J. Liu, X. Cao, Disc-aware ensemble network for glaucoma screening from fundus image, *IEEE Trans. Med. Imaging* 37 (11) (2018) 2493–2501.
- [28] D. Das, D.R. Nayak, R.B. Pachori, CA-Net: a novel cascaded attention-based network for multi-stage glaucoma classification using fundus images, *IEEE Trans. Instrum. Meas.* 72 (2023) 1–10.

- [29] H. Tian, S. Lu, Y. Sun, H. Li, GC-Net: global and class attention blocks for automated glaucoma classification, in: 2022 IEEE 17th Conference on Industrial Electronics and Applications (ICIEA), IEEE, 2022, pp. 498–503.
- [30] D. Das, D.R. Nayak, GS-Net: global self-attention guided CNN for multi-stage glaucoma classification, in: 2023 IEEE International Conference on Image Processing (ICIP), IEEE, 2023, pp. 3454–3458.
- [31] D. Das, D.R. Nayak, R.B. Pachori, AES-Net: an adapter and enhanced self-attention guided network for multi-stage glaucoma classification using fundus images, *Image Vis. Comput.* 146 (2024) 105042.
- [32] A. Geetha, M.C. Sobia, D. Santhi, A. Ahilan, Deep GD: deep learning based snapshot ensemble CNN with EfficientNet for glaucoma detection, *Biomed. Signal Process. Contr.* 100 (2025) 106989.
- [33] M. Rajabpour, H. Kangari, K. Pesudovs, M. Khorrami-Nejad, S. Rahmani, S. Mo-haghegh, S. Moradnejad, Refractive error and vision related quality of life, *BMC Ophthalmol.* 24 (1) (2024) 83.
- [34] P. Bidwai, S. Gite, K. Pahuja, K. Kotecha, A systematic literature review on diabetic retinopathy using an artificial intelligence approach, *Big Data Cognit. Comput.* 6 (4) (2022) 152.
- [35] Z. Wang, Y. Yin, J. Shi, W. Fang, H. Li, X. Wang, Zoom-in-Net: deep mining lesions for diabetic retinopathy detection, in: Medical Image Computing and Computer Assisted Intervention- MICCAI 2017: 20th International Conference, Quebec City, QC, Canada, September 11–13, 2017, Proceedings, Part III 20, Springer, 2017, pp. 267–275.
- [36] N. WANG, C. XIN, J. ZHANG, H. LIU, Outlook on the prevention and treatment of glaucoma in China, *Yan Ke Xue Bao* 36 (6) (2021) 400–404.
- [37] Y. Li, W. Mitchell, T. Elze, N. Zebardast, Association between diabetes, diabetic retinopathy, and glaucoma, *Curr. Diab. Rep.* 21 (2021) 1–16.
- [38] O. Kovalyk, J. Morales-Sánchez, R. Verdú-Monedero, I. Sellés-Navarro, A. Palazón-Cabanes, J.-L. Sancho-Gómez, PAPILA: dataset with fundus images and clinical data of both eyes of the same patient for glaucoma assessment, *Sci. Data* 9 (1) (2022) 291.
- [39] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, S. Xie, A ConvNet for the 2020s, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 11976–11986.
- [40] O. Kovalyk, J. Morales-Sánchez, R. Verdú-Monedero, I. Sellés-Navarro, A. Palazón-Cabanes, J.-L. Sancho-Gómez, Papila Dataset, 2022. <https://figshare.com/articles/dataset/PAPILA>.
- [41] Oia-odir, OIA-ODIR Dataset, 2019. <https://github.com/nkicli/OIA>.
- [42] Origa, ORIGA Dataset, 2010. <https://pubmed.ncbi.nlm.nih.gov/21095735/>.
- [43] Refuge dataset, REFUGE Dataset, 2019. <https://ieee-dataport.org/documents/refuge-retinal-fundus-glaucoma-challenge>.
- [44] J. Wu, H. Fang, F. Li, H. Fu, F. Lin, J. Li, Y. Huang, Q. Yu, S. Song, X. Xu, et al., GAMMA challenge: glaucoma grading from multi-modality images, *Med. Image Anal.* 90 (2023) 102938.
- [45] A. Krizhevsky, I. Sutskever, G.E. Hinton, ImageNet classification with deep convolutional neural networks, *Commun. ACM* 60 (6) (2017) 84–90.
- [46] D.P. Kingma, Adam: A Method for Stochastic Optimization, *arXiv:1412.6980* (2014).
- [47] M. Tan, EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks, *arXiv:1905.11946* (2019).
- [48] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708.
- [49] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
- [50] J. Ha, SMAP: similarity-based matrix factorization framework for inferring miRNA-disease association, *Knowl. Based Syst.* 263 (2023) 110295.
- [51] J. Ha, Deepwalk-based graph embeddings for miRNA–disease association prediction using deep neural network, *Biomedicines* 13 (3) (2025) 536.
- [52] X. Hu, L. Zhang, L. Gao et al., Glim-net: chronic glaucoma forecast transformer for irregularly sampled sequential fundus images, *IEEE Trans. Med. Imaging* 42 (2023) 6, 1875–1884.